

EVA-Flow: Learning Shared Chemistry for Unified Environment-Aware Molecular Conformation Generation

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Abstract

Molecular 3D structures underpin simulations of thermodynamics, kinetics, crystallization, and nucleation, making their accurate prediction central to drug discovery and materials design. However, molecular geometry can change dramatically across environments. Existing models either ignore environmental effects or train separate models for each environment. We hypothesize that transfer learning between different contexts is possible in conformer generation and that a general model will perform better in environments with limited training data: A very common and important setting including ligand binding and crystallization. We introduce *EVA-Flow*, a unified framework for environment-aware molecular conformation generation that learns shared chemical structure and adapts it through environment-conditioned generative modeling. Across vacuum, protein-ligand docking, solvation, and crystal packing, pretraining and cross-environment finetuning provide their largest gains in the three data-scarce settings. Multi-reference Solvation evaluation demonstrates broad ensemble coverage, while expanded same-molecule analyses support environment-specific generation beyond a single canonical geometry. Docking and Packing, for which only one physical reference is available, are evaluated as environment-conditioned structure-prediction tasks.

1 Introduction

Molecules do not exist in a single fixed shape. A drug molecule may be synthesized in solution, crystallized for storage, redissolved in the body, and finally bind a target protein. At each stage the surrounding environment,

whether solvent (Sobornova et al., 2024), crystal lattice (Chattopadhyay et al., 2025; Thompson & Day, 2014; Cruz-Cabeza & Bernstein, 2014), or protein pocket (Gao et al., 2010; Wang & Pang, 2007; Weikl & Paul, 2014), reshapes its 3D geometry.

Despite this, most generative models treat conformer generation as environment-agnostic: molecules are modeled in vacuum (Xu et al., 2022; Hassan et al., 2024) or within a single context such as protein–ligand docking (Corso et al., 2022; 2024), with a separate model for each setting. This fragmentation implicitly assumes that conformational distributions across environments are unrelated tasks. Yet while conformations change across environments, the underlying chemistry does not—the environment merely imposes physical forces that deform a shared molecular structure into different geometries.

This observation motivates a fundamentally different formulation: conformer generation as learning *shared invariant chemistry* under environment-dependent physical constraints, enabling a single model to transfer chemical knowledge across environments via environment-specific conditioning.

We propose *EVA-Flow*, a unified generative framework for environment-aware molecular conformation generation (Figure 1). EVA-Flow couples a variational autoencoder (VAE) (Kingma et al., 2019) encoder with a flow-matching (FM) decoder (Lipman et al., 2022) that generates conformations conditioned on the molecular graph G and environment embedding e . Diagnostics show that the variational latent z is inactive in the reported model, and a latent-free variant is competitive; we therefore treat z as optional rather than attributing molecule–environment information to it.

Our contributions are:

- We formulate conformer generation as learning shared chemistry under environment-specific constraints, rather than training isolated models per environment.
- We introduce a unified environment-conditioned flow-matching architecture that handles diverse environments within a single model; the reported parameterization includes an optional VAE latent and a learned-base training scaffold.
- We demonstrate that pretraining and cross-environment finetuning improve accuracy most clearly in data-scarce environments, and provide expanded same-molecule and multi-reference evidence that the model responds to environmental context. UMA relaxation separately supports physical validity for generated Vacuum conformers (Appendix F).

2 Related Work

Flow matching. Flow matching (FM) trains a continuous normalizing flow by regressing the velocity field that transports a base distribution to the data, yielding a simulation-free objective (Lipman et al., 2022). Rectified flow (Liu et al., 2022) and conditional FM (Tong et al., 2023) extend this to straight-line paths and conditional generation. Equivariant (Hassan et al., 2024) and geometric (Chen & Lipman, 2023) variants adapt FM to 3D molecular settings.

VAE + FM. Hybrid VAE–flow models take two forms: (1) FM in the latent space replaces the VAE prior (Dao et al., 2023), or (2) FM serves as the decoder, mapping a base distribution to data conditioned on a latent code (Fischer et al., 2023; Sargent et al., 2025). We adopt the latter, using an environment-aware FM decoder for conformer generation.

Molecular conformation generation. Classical force-field and MD approaches are accurate but expensive. Neural methods generate conformers via equivariant diffusion (Xu et al., 2022), flows (Hassan et al., 2024), or torsion models (Jing et al., 2022). Most target a single environment—typically vacuum or protein–ligand docking (Corso et al., 2024)—requiring separate models per setting. EVA-Flow addresses this limitation with a single model that conditions on structured environments.

Diffusion in latent space for molecules. Joshi et al. (2025) run diffusion in a learned latent space and decode to coordinates. EVA-Flow also uses a VAE encoder but performs FM in data space and explicitly conditions on the environment.

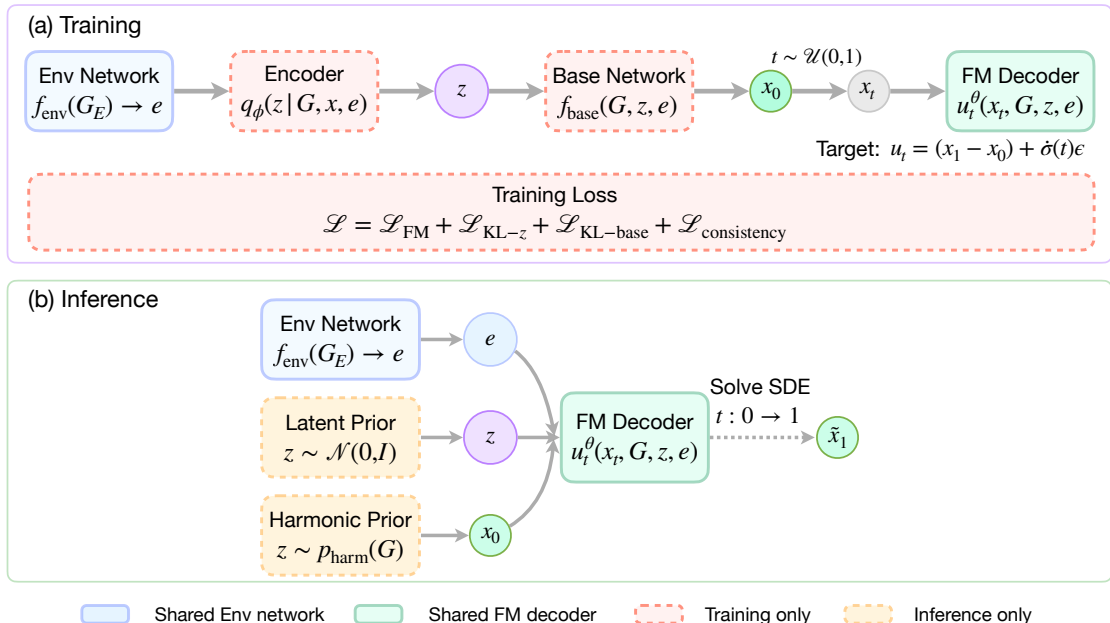


Figure 1: Architecture of EVA-Flow. **(a) Training.** The environment graph G_E is embedded by f_{env} into a vector e . The encoder $q_\phi(z | G, x, e)$ maps the molecular graph G , ground-truth positions x , and environment e to a latent z . The base network $f_{\text{base}}(G, z, e)$ produces a distribution $p_0(x | G, z, e)$ from which x_0 is sampled. The FM decoder learns $u_t^\theta(x_t, G, z, e)$ with stochastic interpolation $x_t = (1-t)x_0 + tx_1 + \sigma(t)\epsilon$, targeting $u_t = (x_1 - x_0) + \sigma(t)\epsilon$. Total loss: $\mathcal{L} = \mathcal{L}_{\text{FM}} + \mathcal{L}_{\text{KL-}z} + \mathcal{L}_{\text{KL-base}} + \mathcal{L}_{\text{consistency}}$. **(b) Inference.** The encoder and base network are bypassed: $e = f_{\text{env}}(G_E)$, $z \sim \mathcal{N}(0, I)$, and $x_0 \sim p_{\text{harm}}(G)$ are sampled independently. All three condition the FM decoder, which integrates $\dot{x}_t = u_t^\theta(x_t, G, z, e)$ from $t=0$ to $t=1$ with an SDE solver. The Env Network and FM Decoder share weights between training and inference.

3 Environment-Aware Flow Matching

3.1 Problem Formulation

Let $G = (V, \mathcal{E})$ denote a molecular graph with nodes representing atoms and edges representing bonds. Each atom is associated with a 3D position $x \in \mathbb{R}^{3N}$, where N is the number of atoms. A molecule exists in an environment E , such as a protein pocket, solvent box, or crystalline lattice. Our general objective is the conditional distribution $p(x|G, E)$, capturing how environments modulate molecular conformations. The available supervision differs by benchmark: Vacuum and Solvation provide multi-conformer references for ensemble evaluation, whereas Docking and Packing provide one physical reference and therefore test environment-conditioned structure prediction. Unlike prior works that train separate models for each E , we seek one model that jointly learns and transfers molecular structure across the environment types observed during training; zero-shot transfer to unseen environment types is outside our scope.

EVA-Flow uses two graph objects with distinct roles. The *target molecular graph* G always describes the molecule whose centered coordinates x are generated; its nodes are molecular atoms and its edges are covalent bonds. The *environment graph* G_E is conditioning context only: it is embedded into a fixed-dimensional vector $e = f_{\text{env}}(G_E)$ and is never generated. Table 1 makes the environment-specific constructions explicit. During training the encoder and learned base additionally access the target coordinates x ; at inference only G , G_E , and their listed attributes are required.

The representation is deliberately heterogeneous. What is shared across environments is the molecular encoder/decoder backbone and the mechanism by which it responds to the conditioning vector e ; the front end that constructs G_E remains environment-specific and incorporates domain knowledge. Our transfer

Table 1: Roles and construction of the target graph G and environment graph G_E . The target graph has the same semantics in every setting; only the context representation changes. ‘‘Coordinates’’ indicates coordinates supplied as input, not the target coordinates to be generated.

Stage/environment	Target G	G_E nodes	G_E edges	Context attributes / coordinates	Available at inference
Pretraining	Molecule atoms/bonds	Random 30–70% atom-induced subgraph	Retained covalent bonds	Atom features and masked-subgraph coordinates	N/A (pretraining only)
Vacuum	Molecule atoms/bonds	Bemis-Murcko scaffold atoms	Scaffold covalent bonds	Atom features; coordinates set to zero	Yes, constructed from input molecule
Docking	Ligand atoms/bonds	Protein-pocket heavy atoms	Atoms within the same residue	Atom features and pocket coordinates	Yes, extracted pocket
Solvation	Solute atoms/bonds	Atoms of a water proxy	Water covalent bonds	sCAV, vCAV, eSOLV, eNELEC; fixed proxy coordinates	Yes, solvent descriptors
Packing	Molecule atoms/bonds	Copy of molecular atoms	Molecular covalent bonds	RDKit geometry, space group, lattice $(a, b, c, \alpha, \beta, \gamma)$	Yes, crystal metadata

claim is therefore shared chemistry across the four observed environment types, not a universal context representation or zero-shot adaptation to a new type.

3.2 Model Architecture

Our framework, EVA-Flow, is a VAE with a FM decoder, consisting of multiple components (Figure 1).

- Environment Embedding Network**(Figure 1). The environment is represented as a graph $G_E = (V_E, \mathcal{E}_E)$, where nodes correspond to atoms with associated features and edges correspond to bonds or interactions. The construction of G_E is described in Section 4.1. The environment network, f_{env} , consists of three graph convolutional network (GCN) layers (Zhang et al., 2019), followed by a two-layer MLP projection to the environment embedding vector $e \in \mathbb{R}^d$:

$$e = f_{\text{env}}(G_E) \quad (1)$$

This embedding provides global information that conditions both the encoder and decoder. The current environment encoder concatenates coordinates to node features before GCN message passing and is therefore not rotation invariant. We chose this common interface because some contexts have physical coordinates (pockets and molecular/crystal proxies) whereas others are descriptor-only; for Docking, we jointly rotate ligand and pocket during training to reduce frame dependence. We nevertheless treat exact context-encoder equivariance as an architectural choice rather than a property of the current GCN, and compare it to an SE(3)-equivariant environment encoder in Appendix N.

- Encoder** (Figure 1). The encoder is a GNN-based module that takes as input the molecular graph G , atomic positions x , and the environment embedding e . Input node features are formed by concatenating the atomic number, node attributes, and atomic positions. These features are processed by SE(3)-equivariant transformer (TorchMD-ET) (Thölke & Fabritiis, 2022; Hassan et al., 2024), producing node embeddings of size $[N, \text{hidden}]$. Each node embedding is then concatenated with the environment embedding e , and projected through two linear layers to produce the mean and log-variance for a node-level latent variational posterior distribution:

$$\begin{aligned} q_\phi(z \mid G, x, e) &= \prod_{i=1}^N \mathcal{N}\left(z_i \mid \mu_{\phi,i}(G, x, e), \text{diag}(\sigma_{\phi,i}^2(G, x, e))\right) \\ &= \mathcal{N}(\mu_\phi(G, x, e), \text{diag}(\sigma_\phi^2(G, x, e))). \end{aligned} \quad (2)$$

where $\mu_{\phi,i}, \sigma_{\phi,i}^2 \in \mathbb{R}^d$ denote the mean and variance predicted for node i . $\mu_\phi, \sigma_\phi^2 \in \mathbb{R}^{N \times d}$ come from stacking all nodes into the appropriate vectorized format. The latent pathway was intended to capture residual molecule–environment variability. However, diagnostics on the reported checkpoint show that its posterior is effectively collapsed, remains unchanged for the same molecule across environments, and can be removed without degrading the matched-budget E4 variant. The decoder therefore primarily uses G and e , and we treat z as optional (Appendix L).

- Base Distribution Network** (Figure 1). We parameterize a conditional Harmonic base distribution using an SE(3)-equivariant transformer (TorchMD-ET), which predicts the parameters of a Harmonic distribution over atomic coordinates. The base distribution is commonly parameterized with zero mean

and the target molecule’s graph Laplacian as precision matrix. We condition this distribution by predicting the mean and diagonal scaling:

$$\mu_{\text{base}}, \lambda_{\text{base}} = f_{\text{base}}(G, x, e, z), \quad (3)$$

where $\mu_{\text{base}} \in \mathbb{R}^{N \times 3}$ is vector-valued and $\lambda_{\text{base}} \in \mathbb{R}^N$ is scalar-valued. Our conditional distribution is then:

$$L_{\text{base}} = L + \text{diag}(\lambda_{\text{base}}), \quad (4)$$

$$\Sigma_{\text{base}} = L_{\text{base}}^\dagger, \quad (5)$$

$$p_0(x_0 \mid G, x, e, z) = \mathcal{N}(x_0 \mid \mu_{\text{base}}, \Sigma_{\text{base}}). \quad (6)$$

where $(\cdot)^\dagger$ is the Moore–Penrose pseudoinverse to handle global translational symmetries. Let $L_{\text{base}} = PDP^\top$ be the eigendecomposition, sampling is performed via:

$$x_0 = \mu_{\text{base}} + PD^{-1/2}\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I) \quad (7)$$

The network is trained by minimizing the KL divergence between the learned distribution and a Harmonic prior defined by the molecular graph topology $\mathcal{N}(\mathbf{0}, L^\dagger)$. During training the network accesses the ground truth x for conditioning. The base network is not used at inference.

4. **FM Decoder** (Figure 1). The decoder generates conformations by transporting samples from the base distribution p_0 to the data distribution using stochastic flow matching. We adopt the TorchMD equivariant network to parameterize a time-dependent vector field u_t^θ that takes as input the current molecular structure x_t , the molecular graph G , the latent variables z , and the environment embedding e :

$$x_0 \sim p'_0(x_0) = \begin{cases} p_0(x_0 \mid G, x, z, e) & \text{Training} \\ \mathcal{N}(x_0 \mid \mathbf{0}, L^\dagger) & \text{Inference} \end{cases} \quad (8)$$

$$\dot{x}_t = u_t^\theta(x_t, G, z, e) \quad (9)$$

We use a stochastic interpolation between base and target samples. Given a base sample $x_0 \sim p'_0$ and a data sample $x_1 \sim p_{\text{data}}$, we define:

$$\begin{aligned} \mu_t &= (1-t)x_0 + tx_1, \\ x_t &= \mu_t + \sigma(t)\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I) \end{aligned} \quad (10)$$

We use the noise schedule:

$$\sigma(t) = \sigma\sqrt{t(1-t)} \quad (11)$$

5. **Inference** (Figure 1). At inference time, given the environment graph G_E , we compute the embedding $e = f_{\text{env}}(G_E)$ and sample a latent $z \sim p(z) = \mathcal{N}(\mathbf{0}, I)$. The initial positions are sampled from the simple Harmonic prior $x_0 \sim \mathcal{N}(\mathbf{0}, L^\dagger)$. Starting at $x_{t_0} = x_0$, we use the implementation’s stochastic Euler sampler: it integrates the learned flow field from $t = 0$ to $t = 1$ and adds zero-center-of-mass Gaussian perturbations at each step. This inference-time perturbation schedule is a sampling heuristic and is distinct from the stochastic interpolation $\sigma\sqrt{t(1-t)}$ used to construct training paths. For $0 = t_0 < t_1 < \dots < t_n = 1$,

$$\begin{aligned} x_{t_{i+1}} &= x_{t_i} + u_\theta(x_{t_i}, t_i; G, z, e) \Delta t_i \\ &\quad + \sqrt{\sigma_{\text{samp}}^2(t_{i+1}) - \sigma_{\text{samp}}^2(t_i)} \xi_i, \\ \xi_i &\sim \mathcal{N}(\mathbf{0}, I), \quad \text{with } \text{COM}(\xi_i) = 0, \end{aligned} \quad (12)$$

where the sampler uses the implemented variance schedule $\sigma_{\text{samp}}^2(t) = t^2$ and $\Delta t_i = t_{i+1} - t_i$. We refer to this stochastic Euler procedure as the “SDE sampler” in the experimental tables for consistency with the implementation, without claiming that it is the probability-flow SDE associated with the training interpolation. The final conformer $\hat{x}$ is obtained after n steps.

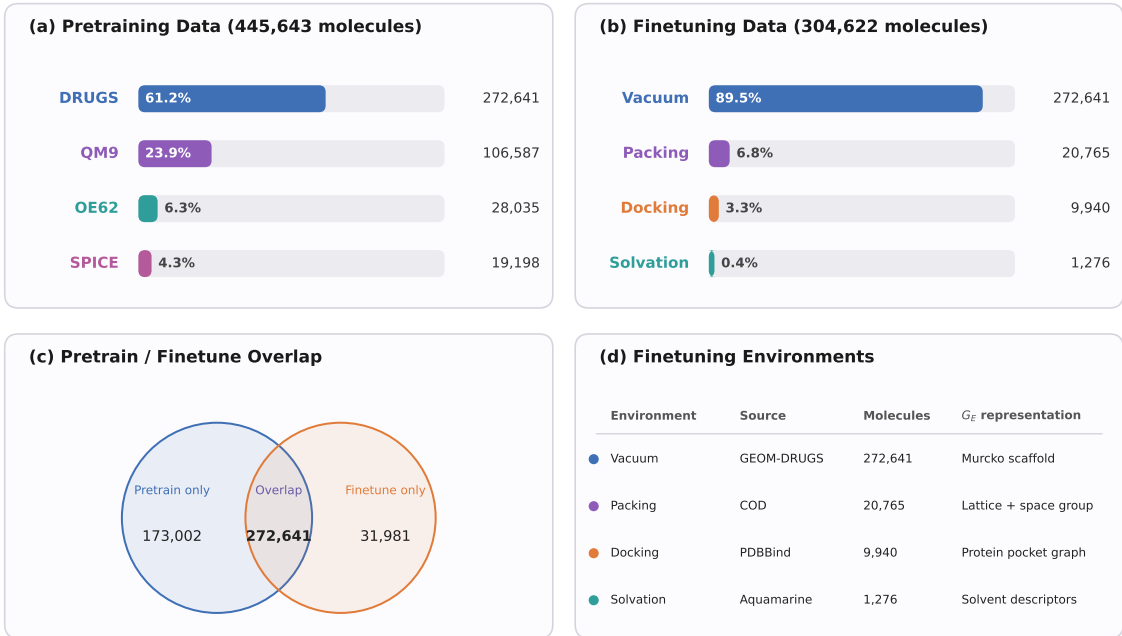


Figure 2: Dataset composition. **(a)** Pretraining corpus by source (445,643 molecules). **(b)** Finetuning corpus by environment (304,622 molecules). **(c)** Overlap between the pretraining and finetuning sets: all 272,641 GEOM-DRUGS molecules used for Vacuum finetuning are also part of the pretraining corpus. **(d)** Data source and environment graph G_E representation for each finetuning environment.

3.3 Training Objectives

We jointly optimize the encoder, environment embedding network, base distribution network, and FM decoder using a composite objective. The total loss consists of four terms, each serving a distinct role (Figure 1):

1. **Latent Prior Regularization.** This is the typical VAE loss to regularize a latent variable z towards a normal:

$$\mathcal{L}_{\text{KL-z}} = D_{\text{KL}}(q_{\phi}(z|x, G, e) \parallel \mathcal{N}(\mathbf{0}, I)). \quad (13)$$

2. **Base Distribution Regularization.** The learned base distribution is regularized by KL divergence toward a simple Harmonic prior induced by graph topology L :

$$\mathcal{L}_{\text{base}} = D_{\text{KL}}(\mathcal{N}(\mu_{\text{base}}, \Sigma_{\text{base}}) \parallel \mathcal{N}(\mathbf{0}, L^{\dagger})). \quad (14)$$

3. **Latent Consistency Loss.** This loss penalizes disagreement between the encoder’s posterior mean given target conformation x and initial conformation $x_0 \sim p'_0(x_0)$. In the reported model the latent collapses, so this term acts as an optimization regularizer rather than a demonstrated carrier of molecule–environment information (Appendix L):

$$\mathcal{L}_{\text{consistency}} = \|\mu_{\phi}(x_0, G, e) - \mu_{\phi}(x, G, e)\|_2^2. \quad (15)$$

4. **FM Loss.** This loss trains the decoder to transport samples from the base distribution to the data distribution through continuous-time dynamics.

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_t} [\|u_t^{\theta}(x_t, G, z, e) - u_t(x_0, x)\|^2], \quad (16)$$

where u_t is the analytically defined target velocity field:

$$\begin{aligned} u_t &= (x_1 - x_0) + \dot{\sigma}(t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I), \\ \dot{\sigma}(t) &= \sigma \frac{1 - 2t}{2\sqrt{t(1-t)}} \end{aligned} \quad (17)$$

All networks are trained jointly at each iteration. We chose to use uniform weighing of the various loss terms:

$$\mathcal{L} = \mathcal{L}_{\text{KL-z}} + \mathcal{L}_{\text{KL-base}} + \mathcal{L}_{\text{consistency}} + \mathcal{L}_{\text{FM}}. \quad (18)$$

These terms regulate two train-inference transitions while preserving coordinate accuracy. $\mathcal{L}_{\text{FM}}$ is the primary geometric objective and learns the coordinate transport. $\mathcal{L}_{\text{KL-z}}$ makes sampling $z \sim \mathcal{N}(\mathbf{0}, I)$ valid at inference; $\mathcal{L}_{\text{KL-base}}$ similarly aligns the learned training-time base with the graph-only harmonic prior used at inference. The consistency term prevents the latent from encoding incidental coordinate-level differences by requiring the same molecule-environment state to have a stable representation at x and x_0 . Thus, the KL and consistency terms constrain information pathways that could otherwise become training-time shortcuts, while the FM term retains the information required for accurate generation. We use unit weights because the implemented losses are normalized per node/batch and have comparable scale; Appendix M reports a one-factor-at-a-time weight sensitivity analysis.

4 Experiments

4.1 Datasets

Pretraining. For large-scale pretraining, we combine datasets containing conformations of small organic and drug-like molecules. This includes relaxed conformations from GEOM-QM9 and GEOM-DRUGS (Axelrod & Gomez-Bombarelli, 2022; Nikitin et al., 2025), as well as higher-energy conformations from the SPICE PubChem subset (Eastman et al., 2023). We further performed relaxation on SPICE dataset using CREST (Pracht et al., 2024) to obtain low-energy conformations. In addition, we include the OE62 dataset (Stuke et al., 2020), which consists of crystal-forming molecules relaxed in vacuum. Altogether, the pretraining corpus contains $\sim 446\text{K}$ molecules and 9.8M conformations (Figure 2a).

During pretraining, we construct G_E by uniformly sampling a mask ratio between 0.3 and 0.7 and retaining the corresponding atom-induced subgraph of the target molecule, including its covalent edges and coordinates. This denoising-style context exposes the shared backbone to diverse partial molecular environments. During Vacuum finetuning and evaluation, we instead apply the RDKit (Landrum & contributors, 2025) MurckoScaffold package (RDKit, 2025) to extract the molecule’s Bemis-Murcko scaffold (Bemis & Murcko, 1996); scaffold coordinates are set to zero so that the context encodes topology rather than the target conformation.

Finetuning. We finetune our model in four environments:

1. **Vacuum.** Molecules in vacuum are isolated. We use the GEOM-DRUGS dataset and construct G_E from the molecule’s Bemis-Murcko scaffold as described above (Figure 2(d)).
2. **Protein-Ligand Docking.** In this task, molecules (ligands) adapt their internal conformation to a known protein binding pocket. We use the PDBBind-v2020 dataset (Liu et al., 2015). The environment graph is constructed using heavy atoms in the extracted protein pocket; edges join atoms in the same residue, and pocket coordinates are included (Figure 2(d)). We center the ligand and predict its pocket-conditioned internal conformation; unlike end-to-end docking systems, EVA-Flow does not predict the ligand’s global translation or orientation in the protein.
3. **Solvation.** Dissolved molecules interact with the solvent. We use the Aquamarine dataset (Medrano Sardonas et al., 2024), which includes conformations relaxed in implicit water. To construct the environment graph, we use a water molecule and add solvation descriptors to the node features such as cavity surface area (sCAV), cavity volume (vCAV), free energy in electrolyte (eSOLV), and non-electrostatic free energy (eNELEC)(Figure 2(d)). These descriptors represent the interaction between molecules and solvent.

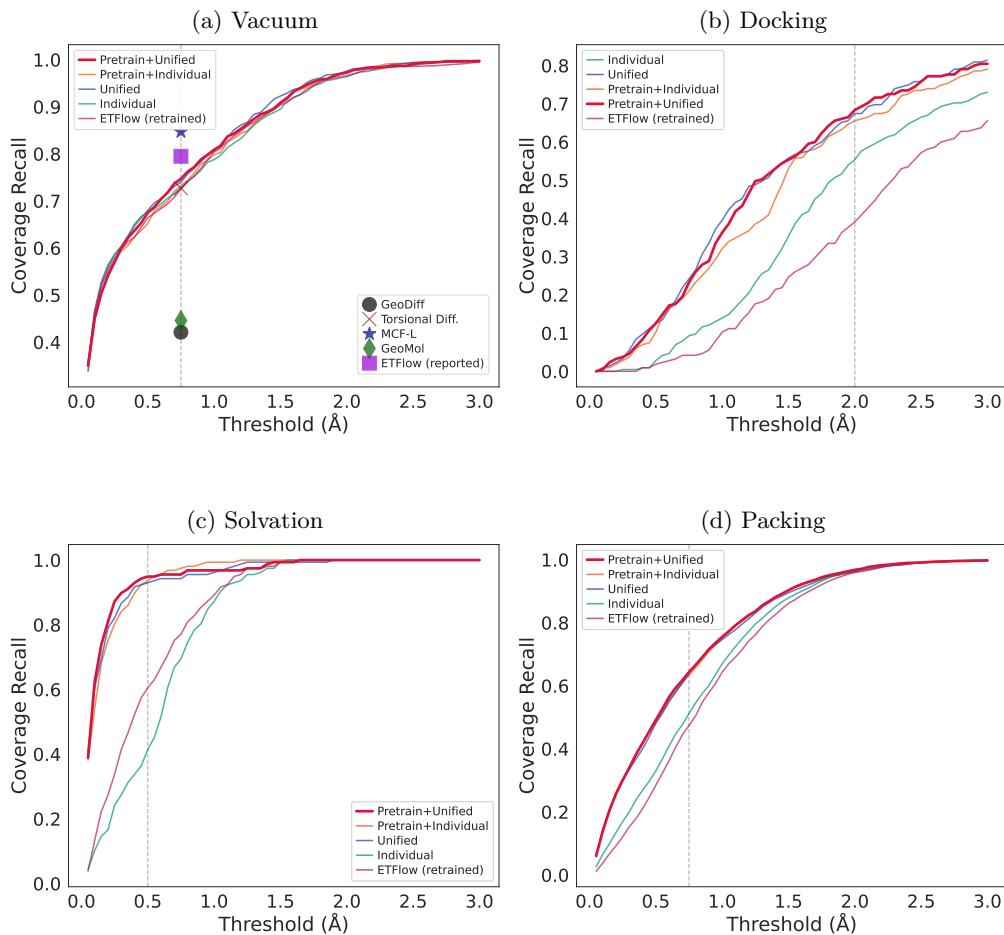


Figure 3: Recall COV as a function of the RMSD threshold across the four environments. In every panel the four EVA-Flow training setups (Individual, Unified, Pretrain+Individual, Pretrain+Unified) are compared to retrained ETFlow. **(a)** Vacuum (cleaned GEOM-DRUGS (Nikitin et al., 2025)); reference points denote published Recall COV from Xu et al. (2022); Ganea et al. (2021); Jing et al. (2022); Wang et al. (2024); Hassan et al. (2024). **(b)** Docking (PDBBind), **(c)** Solvation (Aquamarine), and **(d)** Packing (COD) compare models trained and evaluated under the same centered-conformation protocol. Dashed vertical lines mark the environment-specific RMSD thresholds (0.75/2.0/0.5/0.75 Å).

4. **Crystal Packing.** Molecules interact with neighboring molecules in crystals. We use the Crystallography Open Database (COD) (Gražulis et al., 2009; 2012), and filter to the small organic molecular crystals. We convert the fractional coordinates from the CIF files to Cartesian positions, and recover the bonds with `OpenBabel` (O’Boyle et al., 2008; 2011). For the environment graph, we use a copy of the molecular graph with an RDKit proxy geometry and append the space-group number (sg) and lattice parameters ($a, b, c, \alpha, \beta, \gamma$) to its features (Figure 2(d)).

Further details about data processing and the distributions of number of atoms and the number of rotatable bonds can be found in Sections A and B.

4.2 Experimental Setup

We compare four training strategies.

1. **Pretraining + Individual Finetuning.** We first pretrain the model on the pretraining datasets, then finetune it on a single environment. In the pretraining, the model is exposed to massive molecules and conformations to learn shared chemistry. In the individual finetuning, the model learns a specific environment.
2. **Pretraining + Unified Finetuning.** We first perform training on the pretraining datasets (Section 4.1), but then finetune on all environments of the finetuning datasets. This setup examines if model learns to adapt to different environments.
3. **Unified Finetuning.** We skip the pretraining and only train the model across all environments.
4. **Individual Finetuning.** We train separate models for each environment, without pretraining. This setup is a baseline to study the effects of unification and pretrain.

Evaluation protocols. We use two protocols. The **4-conformer protocol** is our default evaluation across all four environments: for every test molecule we take its single reference conformer and ask each model to generate 4 conformers (Table 2, Figure 3). It is uniform across environments and reflects the practical setting where only one reference geometry is available. Because each molecule contributes one reference and four generations, per-molecule COV can only take values in $\{0, 1\}$ for Recall and $\{0, 0.25, 0.5, 0.75, 1\}$ for Precision; we therefore use mean COV as the primary summary and supplement it with mean/median AMR. Median COV is nevertheless reported for completeness in Appendix E.2. The **2K protocol** generates $2K$ conformers for a molecule with K references. We use it for Vacuum to compare directly with published baselines (Ganea et al., 2021; Jing et al., 2022; Wang et al., 2024; Hassan et al., 2024) (Table 3) and for a separate multi-reference Solvation analysis against all available AQM conformers (Appendix E.4). Recall and Precision must be read jointly because Recall grows with sample budget while Precision penalizes samples that do not match the finite reference set.

Evaluation metrics. We evaluate generated conformers against ground-truth conformers using distance-based RMSD. We report the Recall and Precision Coverage (COV, $\uparrow$) and average minimum RMSD (AMR, $\downarrow$) (Section D). For crystal packing, we also align the generated conformation with the ground-truth one and compute a symmetry-aware RMSD (sRMSD). This value reflects the minimal atomic displacement after accounting for lattice symmetry and periodicity.

4.3 Results

Results under the default 4-conformer protocol are summarized in Table 2 and Figure 3; for a matched comparison we retrain ETFlow with the same task definitions and evaluation code. On Vacuum under the 2K protocol (Table 3), EVA-Flow is comparable to retrained ETFlow (Recall COV 0.721 vs. 0.712; Precision COV 0.707 vs. 0.704) and remains behind the $20\times$ -larger MCF-L on Recall (0.721 vs. 0.847; 12.1M vs. 242M parameters), while exceeding MCF-L on Precision COV (0.707 vs. 0.668). EVA-Flow is therefore not a new state of the art for Vacuum; its accuracy gains are concentrated in the data-scarce Docking, Solvation, and Packing settings. Appendix E.6 separates directly matched comparisons from cross-pipeline reference numbers.

Pretraining improves generalization. Pretraining provides clear gains over Individual training in the data-scarce environments (Table 2, Figure 3). Vacuum differences among EVA-Flow setups are small relative to the observed inference-sampler variability, consistent with its much larger training set.

Unified training outperforms individual models. Unified setups substantially surpass Individual finetuning in Docking, Solvation, and Packing, whereas the differences within Vacuum are small. A data-volume control (total molecule exposure and balanced sampling, Appendix J) supports cross-environment transfer beyond raw target-environment exposure, although the heterogeneous context front ends remain task-specific.

Scope of baseline comparisons. Absolute state-of-the-art comparison is available for Vacuum under the established 2K benchmark, where EVA-Flow is not the best method. End-to-end docking systems additionally predict global translation and orientation and are therefore not matched to our centered internal-conformation task. For Solvation and Packing, no established learned baselines share our task definitions

Table 2: Conformation generation under the **4-conformer protocol** (default) across four environments. Each model generates 4 conformers per molecule, compared with one reference. We report Recall and Precision COV ($\uparrow$, mean) and AMR ($\downarrow$, mean and median) at environment-specific RMSD thresholds: 0.5 Å for Solvation, 0.75 Å for Vacuum and Packing, and 2.0 Å for Docking. For Packing, we additionally report symmetry-aware RMSD (sRMSD, $\downarrow$, mean). Per-molecule COV is quantized, so we report mean COV only. Except for sRMSD, metric cells are mean $\pm$ standard deviation over five independent SDE sampling seeds from one fixed trained checkpoint; this measures inference-sampler variability, not training uncertainty. Best means within each environment are **bolded**.

Environment	Setup	Recall			Precision			sRMSD↓
		COV↑	AMR↓		COV↑	AMR↓		
		mean	mean	median	mean	mean	median	
Vacuum	Pretrain+Unified	0.747±0.013	0.468±0.012	0.144±0.024	0.463±0.004	0.969±0.006	0.900±0.023	–
	Pretrain+Individual	0.733±0.010	0.481±0.012	0.146±0.019	0.454±0.004	0.980±0.008	0.925±0.016	–
	Unified	0.740±0.007	0.456±0.008	0.131±0.011	0.463±0.003	0.954±0.005	0.879±0.011	–
	Individual	0.728±0.005	0.487±0.004	0.134±0.006	0.460±0.003	0.978±0.004	0.911±0.017	–
	ETFlow (retrained)	0.725±0.013	0.488±0.012	0.130±0.024	0.467±0.004	0.972±0.006	0.919±0.023	–
Docking	Pretrain+Unified	0.684±0.013	1.899±0.027	1.257±0.037	0.512±0.008	2.471±0.023	1.864±0.039	–
	Pretrain+Individual	0.656±0.010	1.980±0.013	1.463±0.021	0.458±0.006	2.581±0.019	2.076±0.034	–
	Unified	0.674±0.021	1.889±0.016	1.303±0.060	0.536±0.012	2.391±0.021	1.732±0.023	–
	Individual	0.553±0.013	2.458±0.034	1.833±0.039	0.378±0.009	3.135±0.032	2.387±0.075	–
	ETFlow (retrained)	0.391±0.013	2.768±0.027	2.288±0.037	0.242±0.008	3.580±0.023	2.958±0.039	–
Solvation	Pretrain+Unified	0.949±0.019	0.153±0.013	0.072±0.016	0.780±0.009	0.367±0.007	0.232±0.013	–
	Pretrain+Individual	0.936±0.018	0.157±0.006	0.087±0.020	0.693±0.006	0.431±0.007	0.334±0.017	–
	Unified	0.930±0.034	0.166±0.017	0.083±0.018	0.769±0.014	0.370±0.009	0.249±0.023	–
	Individual	0.414±0.021	0.580±0.020	0.568±0.026	0.177±0.009	0.961±0.045	0.851±0.044	–
	ETFlow (retrained)	0.605±0.019	0.475±0.013	0.366±0.016	0.264±0.009	0.824±0.007	0.786±0.013	–
Packing	Pretrain+Unified	0.644±0.004	0.664±0.002	0.509±0.005	0.410±0.001	1.073±0.001	1.008±0.005	0.055
	Pretrain+Individual	0.633±0.005	0.672±0.003	0.529±0.006	0.400±0.002	1.087±0.001	1.022±0.005	0.055
	Unified	0.640±0.003	0.682±0.005	0.527±0.006	0.414±0.002	1.067±0.002	0.997±0.008	0.054
	Individual	0.512±0.006	0.819±0.004	0.734±0.008	0.303±0.002	1.222±0.004	1.161±0.004	0.064
	ETFlow (retrained)	0.474±0.004	0.878±0.002	0.794±0.005	0.274±0.001	1.274±0.001	1.203±0.005	0.069

and splits; we report retrained ETFlow under identical evaluation and add ETKDG+MMFF and GFN2-xTB physical references in Appendix E.5. These references contextualize performance but do not constitute controlled ablations of environment conditioning. For Docking, retrained ETFlow is selected by validation loss after 100 epochs on the same PDBBind split, molecular inputs, sample budget, and evaluator. ETFlow has no published internal-conformation result under this protocol. This is a matched ligand-only reference, but a shared schedule does not guarantee architecture-specific optimal tuning; the inference-time NoPocket ablation (Appendix H) more directly tests whether EVA-Flow uses pocket information.

4.4 Analysis

To test whether EVA-Flow responds to the environment rather than always generating one geometry, we match shared molecules by canonical SMILES across all six environment pairs (Table 5; matching and additional controls are detailed in Appendix G). For a pair A/B , “own A/B ” is the mean heavy-atom RMSD from generations conditioned on A or B to that same environment’s reference; “cross A/B ” instead uses the other environment’s reference; and “shift” is the RMSD between the two conditioned generations. The own-versus-cross pattern is strongest when the reference geometries are well separated (e.g., Solvation–Vacuum) and weak for Packing–Docking, whose references are similar.

We additionally retrain after removing all 1,201 molecules appearing in at least two environments from every finetuning dataset. The own-versus-cross pattern persists for the well-separated pairs (Appendix G), but own-reference error increases. Because some molecules may occur in the broad Vacuum pretraining corpus, this is a *finetuning-held-out* rather than fully pretraining-decontaminated control. It reduces, but does not eliminate, the concern that the result arises solely from memorization. Zeroing domain-specific context features also degrades each environment (Appendix H). Pairwise RMSD among generated samples establishes non-trivial spread (Appendix E.3) but is not itself evidence of calibrated distributional probabilities.

Table 3: Conformation generation for Vacuum under the **2K protocol**: each model generates $2K$ conformers per molecule, where K is the number of reference conformers. We report mean and median Recall and Precision COV ($\uparrow$) and AMR ($\downarrow$) at RMSD threshold $0.75 \text{ \AA}$. Best results are **bolded**. The models are trained on the original GEOM-DRUGS dataset and evaluated on its full test set. Literature numbers for GeoDiff, GeoMol, Torsional Diffusion, MCF-L, and ETFlow (reported) are taken from Table 1 of Hassan et al. (2024); ETFlow (reported) refers to the ET-Flow base (8.3M) variant. ETFlow (retrained) and EVA-Flow are trained on all available conformers per molecule on the original GEOM-DRUGS and evaluated on the same test set.

Setup	Recall				Precision			
	COV $\uparrow$		AMR $\downarrow$		COV $\uparrow$		AMR $\downarrow$	
	mean	median	mean	median	mean	median	mean	median
GeoDiff	0.421	0.378	0.835	0.809	0.249	0.145	1.136	1.090
GeoMol	0.446	0.414	0.875	0.834	0.430	0.364	0.928	0.841
Torsional Diff.	0.727	0.800	0.582	0.565	0.552	0.569	0.778	0.729
MCF-L	0.847	0.922	0.390	0.247	0.668	0.713	0.618	0.530
ETFlow (reported)	0.795	0.846	0.452	0.419	0.744	0.810	0.541	0.470
ETFlow (retrained)	0.712	0.742	0.562	0.530	0.704	0.758	0.594	0.520
EVA-Flow (Pretrain+Unified)	0.721	0.756	0.550	0.527	0.707	0.781	0.589	0.510

Table 4: Visualization of reference and generation conformations in Pretrain+Unified setup for shared molecules in Solvation-Packing and Solvation-Docking. The conformers are aligned to the reference of Solvation.

Solvation-Packing				Solvation-Docking			
ref_solv	gen_solv	ref_pack	gen_pack	ref_solv	gen_solv	ref_dock	gen_dock
							
							

Table 5: Expanded shared-molecule analysis (Pretrain+Unified). All values are mean heavy-atom RMSD in $\text{\AA}$.

Pair (A/B)	n	own A	own B	cross A	cross B	generation shift
Solvation–Vacuum	142	0.12	0.64	3.41	3.31	3.41
Docking–Vacuum	558	0.75	0.76	1.22	1.21	1.20
Vacuum–Packing	487	0.33	0.35	0.64	0.65	0.64
Docking–Solvation	13	0.55	0.09	2.41	2.49	2.42
Packing–Solvation	23	0.06	0.09	1.49	1.48	1.48
Packing–Docking	64	0.28	0.36	0.45	0.38	0.42

When environments agree, generations agree. Packing–Docking is the weakly separated case: its reference and generated cross-environment structures are already similar. This complements the divergent well-separated pairs and is consistent with the model responding to context rather than enforcing different outputs solely from the environment label.

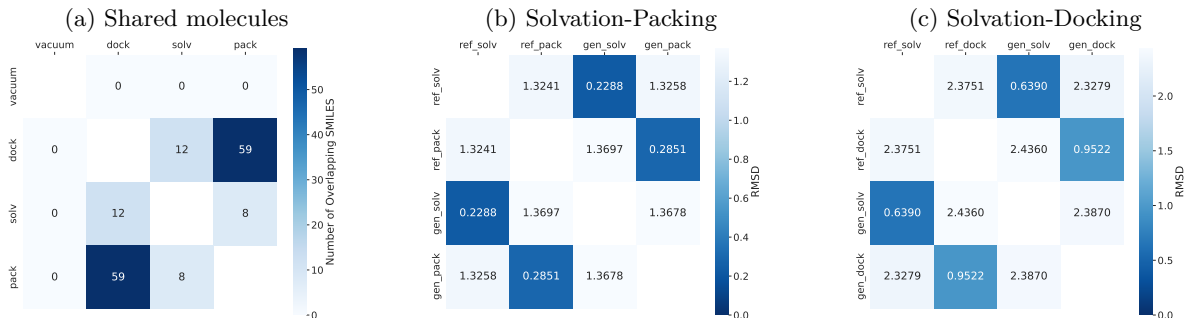


Figure 4: Shared-molecule analysis across environments. (a) Pairwise counts of molecules shared between environments. Darker cells indicate more molecules. (b, c) Pairwise heavy-atom RMSD (Å) between reference and generated conformers for molecules shared across Solvation-Packing and Solvation-Docking. We report the mean RMSD after alignment for: $\text{ref_A} \leftrightarrow \text{ref_B}$, $\text{gen_A} \leftrightarrow \text{ref_A}$, $\text{gen_A} \leftrightarrow \text{ref_B}$, $\text{gen_B} \leftrightarrow \text{ref_A}$, $\text{gen_B} \leftrightarrow \text{ref_B}$, and $\text{gen_A} \leftrightarrow \text{gen_B}$ (A/B = the two environments). Darker cells indicate lower RMSD (better agreement).

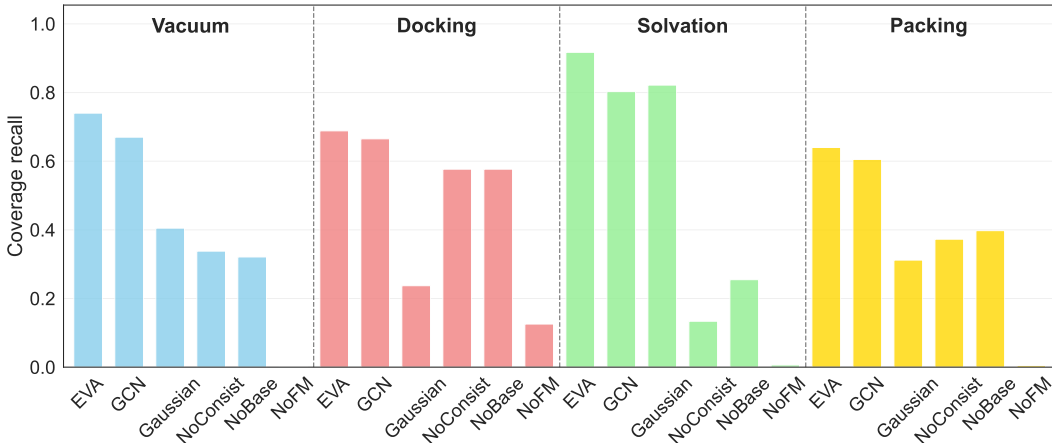


Figure 5: Recall COV for all environments (Vacuum, Docking, Solvation, Packing) using the Unified setup for small model. ‘EVA’ refers to standard setup; ‘GCN’ uses non-equivariant networks for the encoder and base; ‘Gaussian’ parameterizes the base distribution as a Gaussian distribution instead of Harmonic; ‘NoConsist’ is removing latent consistency loss; ‘NoBase’ is for replacing the learned base distribution network for initial positions x_0 with a Harmonic distribution determined only by the graph; ‘NoFM’ is removing the FM decoder and using a regression decoder.

4.5 Ablation Studies

We ablate key components using the Unified setup with the small model (Figure 5). The full model (EVA) is defined in Sections 3.2 and 3.3. Variants: **GCN** replaces SE(3)-equivariant modules with graph convolutional networks; **Gaussian** uses a diagonal Gaussian base instead of the Harmonic distribution; **NoConsist** removes the latent consistency loss; **NoBase** replaces the learned base network with the graph-topology Harmonic prior $\mathcal{N}(\mathbf{0}, L^\dagger)$; **NoFM** replaces the FM decoder with a regression decoder; and **E4** coherently removes both the learned base and latent, retaining environment conditioning, the fixed harmonic base, and the FM decoder.

Figure 5 reports Recall COV of the original ablations; complete metrics and the E4 result are reported in Appendix I. E4 differs from the full model by +0.011/ +0.042/ +0.050/ −0.007 Recall COV on Vacuum/Docking/Solvation/Packing, showing that the latent-free conditional flow is competitive. The ablation of the FM decoder consistently underperforms across all environments, because regression tends to learn the conditional mean of a multi-modal conformer distribution, leading to blurred geometry and lower coverage.

Removing only the learned base at training (NoBase) causes a large drop, yet the learned base nearly coincides with the graph-only harmonic prior at convergence (Appendix K). The NoBase latent activates while the full-model latent is collapsed, an optimization interaction consistent with the train/inference mismatch observed for NoBase. This is a diagnostic interpretation rather than a direct causal intervention. E4 shows that the learned base is useful within the original coupled base-latent parameterization but is not necessary when both pathways are removed coherently. Eliminating the latent consistency loss (NoConsist) reduces Recall COV in the original parameterization, suggesting an optimization-regularization role rather than information carried by the collapsed latent. Replacing the harmonic base with a Gaussian one (Gaussian) also degrades performance, highlighting the importance of incorporating graph geometry through the harmonic formulation. Finally, using non-equivariant encoders and base networks (GCN) leads to inferior results, demonstrating the benefit of SE(3)-equivariant architectures for conformer generation.

5 Conclusion

We introduced EVA-Flow, a unified framework for environment-aware molecular conformation generation. Its shared flow-matching backbone conditions on heterogeneous representations of vacuum, protein pockets, solvation descriptors, and crystal metadata. The largest gains from pretraining and unified finetuning occur in the three data-scarce environments. Multi-reference Solvation evaluation supports broad ensemble coverage, while expanded shared-molecule and context-ablation analyses support environment-responsive generation; Docking and Packing remain single-reference structure-prediction evaluations.

Several directions remain open. The variational latent collapses, and a simpler latent-free E4 model is competitive, motivating a leaner default architecture. The loss-weight study varies one component at a time and does not cover width/depth or interactions. More expressive backbones and scaling could improve accuracy, while multi-reference benchmarks and environment-specific physical validation are needed beyond Solvation and Vacuum. Extending to unseen environment types such as surface adsorption or solid-liquid interfaces remains future work.

Broader Impact Statement

EVA-Flow is intended to support molecular modeling in chemistry, materials science, and drug discovery. More accurate conformation generation could reduce the computational cost of downstream simulations and facilitate scientific discovery. Like other molecular generative and predictive tools, however, the method may produce inaccurate geometries and should not replace experimental validation or high-accuracy physical simulation in safety-critical applications. We recommend that generated structures be validated with appropriate physical calculations and domain expertise before downstream use.

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A Data Preparation

GEOM QM9 and DRUGS. We follow the data processing pipeline of ETFlow (Hassan et al., 2024) and adopt the same train/validation/test splits. For GEOM-DRUGS, We applied the data cleaning strategies reported in (Nikitin et al., 2025) that fixes valency, bond-order, and force-field inconsistencies in the original datasets. For each molecule, we retain the top 30 conformers ranked by Boltzmann weight. Each molecule is stored as a PyTorch Geometric object containing atomic numbers, 3D positions, and a canonical SMILES string. The QM9 training set contains 106,587 molecules (794,960 conformations) and the validation set 13,323 molecules (99,945 conformations); the test set has 1,000 molecules. The DRUGS training set contains 272,641 molecules (5,770,377 conformations) and the validation set 30,405 molecules (715,592 conformations); the test set has 1,000 molecules.

SPICE and CREST-relaxed SPICE. We use the PubChem subset of SPICE (Eastman et al., 2023), which contains small drug-like molecules (18–50 atoms; elements Br, C, Cl, F, H, I, N, O, P, S).

Because the SPICE conformers are not fully relaxed, we construct a complementary relaxed set via CREST (Pracht et al., 2024). For each molecule we (i) generate 50 initial conformers with RDKit `EmbedMultipleConfs` (pruning threshold 0.01 Å, MMFF optimization), (ii) deduplicate with `GetBestRMS` (threshold 0.1 Å), (iii) refine the ten lowest-energy conformers with extended tight binding (xTB) (Grimme et al., 2017; Friede et al., 2024), and (iv) run CREST v3.0.2 from the lowest-xTB-energy conformer with a 6 kcal/mol cutoff. Molecules with cis/trans stereo-SMILES were excluded because the stereochemistry is not preserved through the pipeline.

For both SPICE and CREST-relaxed SPICE, we align atom ordering between the dataset coordinates and the RDKit molecule built from the canonical SMILES using `GetSubstructMatch`. We split 0.8/0.1/0.1 into train/validation/test. The training set contains 19,198 molecules (959,802 SPICE conformers; 2,203,392 CREST conformers), the validation set 2,508 molecules (125,370; 278,673), and the test set 2,941 molecules.

OE62. OE62 (Stuke et al., 2020) provides PBE-relaxed geometries of crystal-forming organic molecules. We read the SMILES and coordinates, align atom ordering via `GetSubstructMatch` as above, and split 0.8/0.1/0.1. The training set contains 28,038 molecules, the validation set 4,297, and the test set 3,933 (one conformation each).

PDBBind. We use PDBBind-v2020 (Liu et al., 2015). For each protein–ligand pair, we extract the ligand from the SDF file, add hydrogens, and generate a canonical SMILES. Atom ordering between the SDF-derived and SMILES-derived RDKit molecules is aligned via `GetSubstructMatch`.

The protein binding pocket serves as the environment graph. We parse the pocket PDB file and retain only heavy atoms, adding edges between atoms within the same residue. Node features follow ETFlow conventions.

We use the original PDBBind splits. After filtering invalid ligands, the training set has 9,940 examples, the validation set 614, and the test set 218 (one ligand conformation each).

Aquamarine (AQM). Aquamarine (Medrano Sandonas et al., 2024) contains conformations relaxed in implicit water. We build RDKit molecules from SMILES, add hydrogens, and align atom ordering as above. The environment graph uses a fixed RDKit-embedded water proxy with four normalized solvation descriptors appended to node features: cavity surface area (sCAV), cavity volume (vCAV), free energy in electrolyte (eSOLV), and non-electrostatic free energy (eNELEC).

After filtering, we split molecules 0.8/0.1/0.1. The processed files retain all Aquamarine conformers, but the default one-reference protocol selects the dataset’s `c1` (index-0) conformer for training and evaluation, yielding 1,276 training, 175 validation, and 157 test molecules. This is a fixed canonical-index choice rather than an energy ranking; we report a conformer-selection sensitivity analysis in Appendix O.

Crystallography Open Database (COD). The COD (Grazulis et al., 2009; 2012) is an open-access collection of crystal structures. We select organic molecular crystals by applying the following filters: the structure must (i) contain organic molecules, (ii) not be marked as disordered (Groom et al., 2016), (iii) not be missing hydrogen, (iv) have a valid SMILES (Weininger, 1988) with a single fragment, (v) pass structural validation (Ong et al., 2013), and (vi) contain exactly one formula unit in the asymmetric unit ($Z' = 1$).

We recover the molecular topology with RDKit maximum-common-substructure matching (`FindMCS`) (Lan-drum & contributors, 2025) rather than the often-unreliable COD SMILES, and extract the space group, Wyckoff positions, and Cartesian coordinates for each formula unit using `pymatgen` (Ong et al., 2013). We process structures into PyTorch Geometric objects, aligning atom ordering via RDKit `GetSubstructMatch`. Lattice parameters ($a, b, c, \alpha, \beta, \gamma$) and the space group number are included as features.

After filtering, we split 0.8/0.1/0.1. The training set has 20,765 molecules, the validation set 3,239, and the test set 3,235 (one conformation each).

B Distribution of Number of Atoms and Number of Rotatable Bonds

To characterize the intrinsic difficulty of conformer generation across datasets, we report histograms of molecular size (number of atoms, N_{atoms}) and flexibility (number of rotatable bonds, N_{rot}) for each dataset (Figures 6 and 7). We compute N_{rot} using `rdMolDescriptors.CalcNumRotatableBonds` from RDKit with its default definition (non-ring single bonds between non-terminal heavy atoms).

Across the pretraining sets (GEOM Drugs, GEOM QM9, SPICE, CREST-relaxed SPICE, and OE62), N_{atoms} concentrates in the mid-size regime with means in the ~ 17 –44 range, while N_{rot} is typically modest (most molecules have fewer than 15 rotatable bonds).

For the finetuning sets (GEOM Drugs, PDBBind, AQM, and COD), the atom-count distribution shifts larger with means in the ~ 38 – 64 range. In particular, the PDBBind dataset exhibits a long tail (very large molecules), indicating increased geometric and torsional complexity.

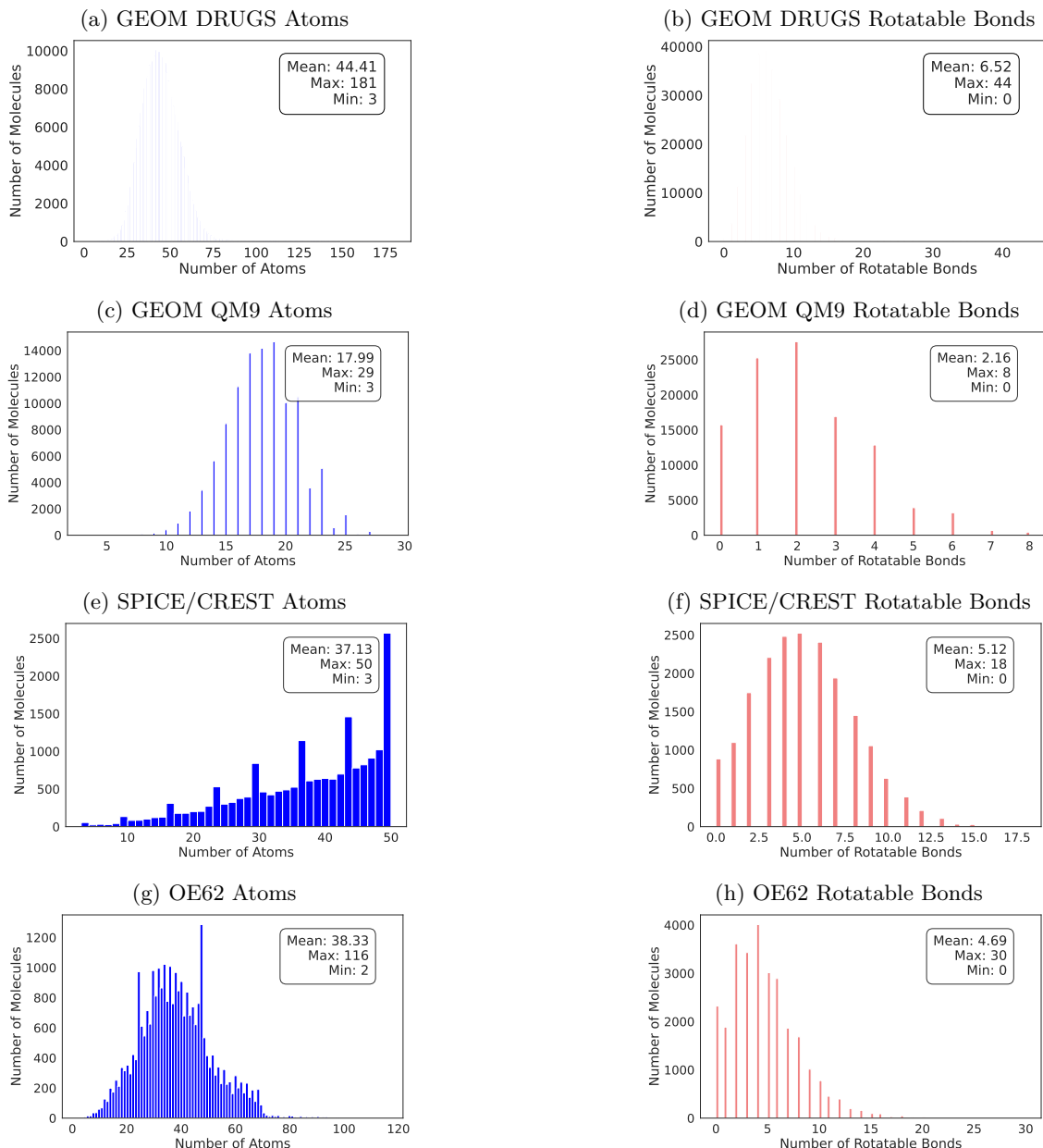


Figure 6: The histograms of the number of atoms and the number of rotatable bonds in the pretraining datasets, including GEOM DRUGS, GEOM QM9, SPICE, CREST-relaxed SPICE, and OE62.

C Training Details and Hyperparameters

All models are trained on 8 NVIDIA A100 GPUs with AdamW and a cosine-annealing learning rate schedule with linear warmup. Table 6 lists the per-size hyperparameters.

Pretraining. We train for 800 epochs, resampling 100K examples per epoch from the pretraining pool with mixing probabilities 0.6 (GEOM-DRUGS), 0.2 (OE62), 0.1 (GEOM-QM9), 0.05 (SPICE), and 0.05 (CREST).

Unified finetuning. Starting from the pretrained checkpoint, we finetune for 800 epochs using nominal allocations from a 100K-example epoch: 0.6 (Vacuum / GEOM-DRUGS), 0.2 (Packing / COD), 0.15 (Dock-

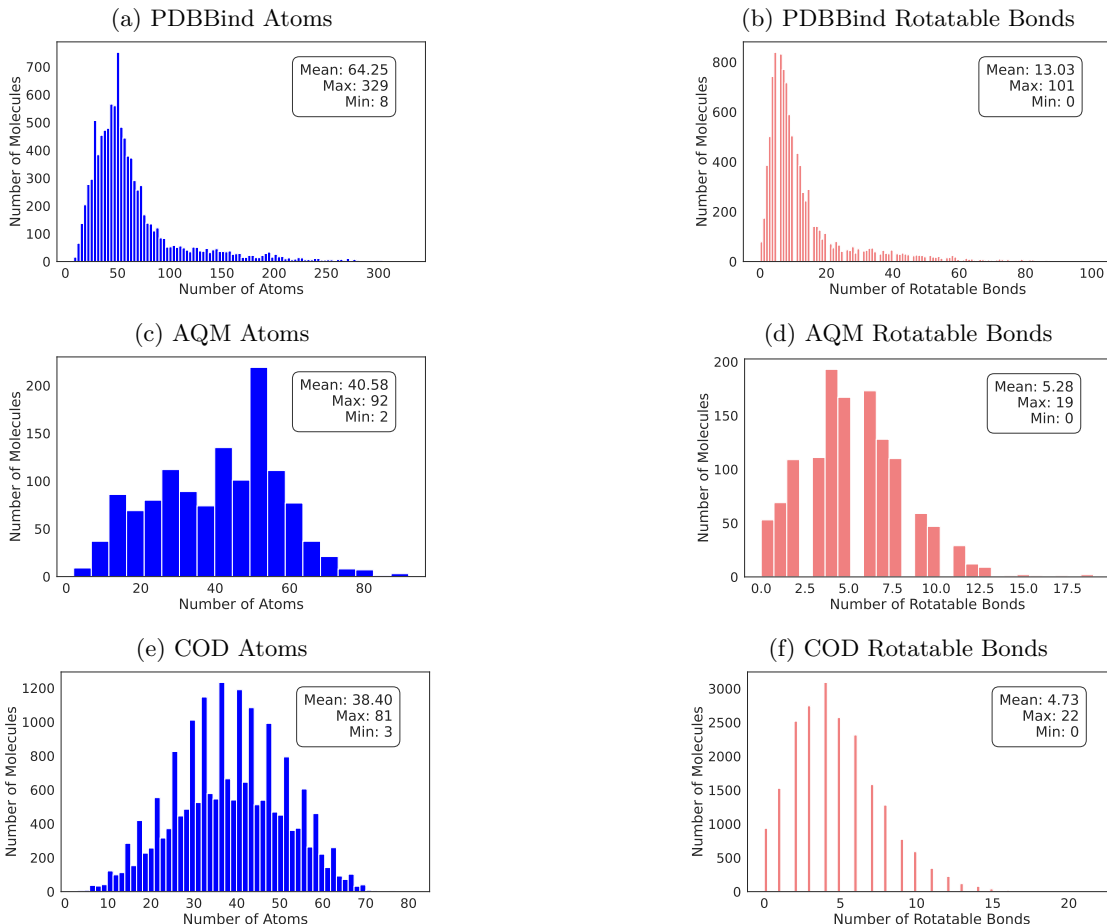


Figure 7: The histograms of the number of atoms and the number of rotatable bonds in the finetuning datasets, including PDBBind, AQM, and COD. The GEOM DRUGS dataset is the same as the one in the pretraining datasets (Figure 6).

ing / PDBBind), and 0.05 (Solvation / AQM). Each allocation is capped at the available dataset size, giving 91,216 effective samples per epoch.

Individual finetuning. We finetune on each environment’s full training set for 800 epochs using the same optimizer and schedule.

C.1 Training, Inference, and Reproducibility Details

Optimization and schedule. We use AdamW with $(\beta_1, \beta_2) = (0.95, 0.999)$, weight decay 10^{-8} , and gradient-norm clipping at 1.0. The learning rate follows a cosine-annealing-with-warmup schedule: a linear warmup of 31,250 steps to a peak of 5×10^{-4} , then cosine decay over a 312,500-step cycle to a minimum of 10^{-8} .

Data splits, validation, and checkpointing. Each environment uses fixed train/validation/test splits. We validate every epoch and monitor the total validation loss (`val/total_loss`); the model checkpoint with the lowest validation loss is used for all reported results.

Random seeds. Training is seeded with seed 42. Inference uses a stochastic SDE sampler. Table 2 reports mean $\pm$ standard deviation over five independent sampling seeds from each fixed checkpoint; these

Table 6: Hyperparameters for EVA-Flow.

Hyperparameter	Value
Batch size	32
FM decoder layers	20
Hidden channels	160
Attention heads	8
Latent dim (d_z)	128
Environment dim (d_e)	128
Encoder layers	3
Base layers	3
Environment layers	3
Cutoff range	[0.0, 10.0] Å
Activation	SiLU
RBF type / count	ExpNorm / 64
Optimizer	AdamW
AdamW (β_1, β_2)	(0.95, 0.999)
Max LR	5×10^{-4}
Min LR	10^{-8}
Weight decay	10^{-8}
Warmup steps	31,250
Cosine cycle length	312,500 steps
Gradient clip (norm)	1.0
Random seed	42
Inference sampler / steps	SDE / 50
SDE noise (s_{churn} , std)	(1.0, 1.0)
GPUs	8× A100
Total parameters	12.1M

Table 7: Per-molecule inference time on a single A100 GPU (GEOM-DRUGS, 1,000 molecules). EVA-Flow uses the Pretrain+Unified setup.

Method	Sampler	Steps	R-COV $\uparrow$	Time (s/mol) $\downarrow$
ETFlow (retrained)	SDE	50	0.725	0.309
EVA-Flow (P+U)	SDE	50	0.747	0.266
EVA-Flow (P+U)	SDE	100	0.752	0.530

values quantify inference-sampler variability, not training uncertainty. The separate all-conformer ETFlow diagnostic uses six runs.

Inference sampler and noise. Unless otherwise noted, all reported numbers use the SDE sampler with 50 integration steps, additive-noise scale $\text{std} = 1.0$, churn $s_{\text{churn}} = 1.0$, and a stochastic window $t \in [10^{-4}, 0.9999]$; the initial state x_0 is drawn from the harmonic prior with no additional perturbation. The deterministic ODE sampler is also supported and trades a small amount of Recall COV for speed (Vacuum: 0.737 ODE vs. 0.747 SDE).

Inference time. Table 7 reports per-molecule inference time on a single A100 GPU (GEOM-DRUGS test set). EVA-Flow is modestly faster than retrained ETFlow at the same step count in our implementation; both use their respective graph-conditioned vector fields and the fixed harmonic initialization.

Compute. Models are trained on 8× NVIDIA A100 GPUs. Each 800-epoch stage (pretraining and unified finetuning) takes roughly 5 minutes per epoch, i.e. approximately 67 wall-clock hours (~ 530 A100-GPU-hours); individual finetuning is run per environment on its full training set.

Dataset preprocessing. Each dataset is processed by a dedicated script that will be released in the future. Molecular graphs and features are built with `RDKit` and `datamol`; crystal lattices and space groups for Packing use `pymatgen`, with `OpenBabel` for bond perception from CIF files; numerical processing uses `NumPy` and `SciPy`.

D Evaluation Metric Definition

We compute average minimum RMSD (AMR) and Coverage (COV) Recall and Precision to assess the performance of molecular conformer generation following the approaches of (Ganea et al., 2021; Xu et al., 2022; Jing et al., 2022). Recall measures the extent to which the generated conformers capture the ground-truth conformers, while Precision indicates the proportion of generated conformers that match the ground-truth conformers. Using C_g to denote the set of generated conformations, and C_r the set of reference conformations, we calculate these metrics using the following equations:

$$\text{AMR-R}(C_g, C_r) = \frac{1}{|C_r|} \sum_{\mathbf{R} \in C_r} \min_{\hat{\mathbf{R}} \in C_g} \text{RMSD}(\mathbf{R}, \hat{\mathbf{R}}) \quad (19)$$

$$\text{COV-R}(C_g, C_r) = \frac{1}{|C_r|} |\{\mathbf{R} \in C_r | \text{RMSD}(\mathbf{R}, \hat{\mathbf{R}}) < \delta, \hat{\mathbf{R}} \in C_g\}| \quad (20)$$

$$\text{AMR-P}(C_r, C_g) = \frac{1}{|C_g|} \sum_{\hat{\mathbf{R}} \in C_g} \min_{\mathbf{R} \in C_r} \text{RMSD}(\hat{\mathbf{R}}, \mathbf{R}) \quad (21)$$

$$\text{COV-P}(C_r, C_g) = \frac{1}{|C_g|} |\{\hat{\mathbf{R}} \in C_g | \text{RMSD}(\hat{\mathbf{R}}, \mathbf{R}) < \delta, \mathbf{R} \in C_r\}| \quad (22)$$

A lower AMR signifies closer matches. Recall COV measures reference coverage, whereas Precision COV measures the fraction of generated samples close to a finite reference set; neither alone establishes sample diversity or calibrated probabilities. The threshold δ is set to 0.5 Å for Solvation, 0.75 Å for Vacuum (Hassan et al., 2024) and Packing, and 2.0 Å for Docking (Corso et al., 2022).

E Additional Results

E.1 Additional Precision Results

Precision Coverage plots are shown in Figures 8.

E.2 Median Coverage under the 4-Conformer Protocol

For completeness, Table 8 reports median Coverage. With one reference and four generated conformers per molecule, per-molecule Recall COV is binary and Precision COV is restricted to $\{0, 0.25, 0.5, 0.75, 1\}$. Consequently, the median values are strongly quantized and often identical across models; mean COV and median AMR in Table 2 provide substantially finer discrimination.

E.3 Generation Diversity

To test for sample collapse, we measure the mean symmetry-aware heavy-atom best-RMS over all unique pairs of the four generated conformers per molecule, averaged over the test set (Pretrain+Unified model). Table 9 reports non-zero spread in every environment. This establishes sample diversity but does not determine whether the generated probability density matches the physical ensemble.

E.4 Multi-Reference Solvation Evaluation

Aquamarine test molecules contain a mean of 37 conformers (up to 431), although the default training and 4-conformer evaluation use `c1`. We therefore evaluate against all available references with the 2K protocol

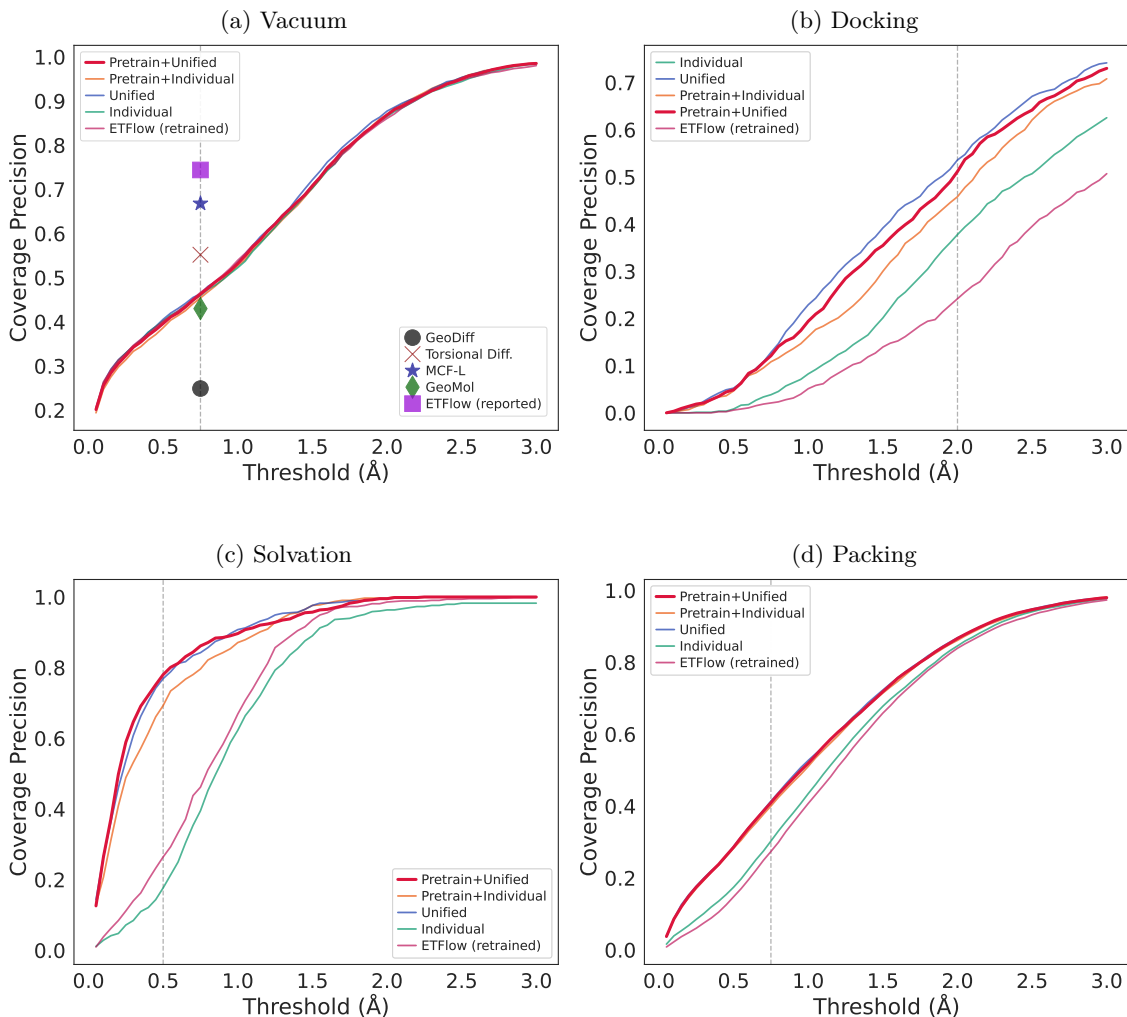


Figure 8: Precision Coverage for all four environments: Vacuum, Docking, Solvation, and Packing.

(Table 10). Recall and Precision are reported jointly: high Recall demonstrates broad coverage of the available ensemble, while Precision penalizes generations that do not match that finite reference set. The result does not by itself establish calibrated physical probabilities.

E.5 Physical Reference Methods

We add two non-learned reference methods under the same four-sample evaluation budget: RDKit ETKDG followed by MMFF relaxation for all environments, and GFN2-xTB with ALPB-water implicit solvation for Solvation. These are physical reference points rather than controlled ablations of environment conditioning because they differ from EVA-Flow in model class, data, and optimization.

E.6 Reconciling Published and Retrained ETFlow Results

The ETFlow number reported in its paper uses the original GEOM-DRUGS benchmark, all available reference conformers, and a 2K sampling budget. This differs from our default Table 2, which uses the chemically cleaned benchmark, one reference, and four samples. Table 12 evaluates the published checkpoint and retrained ETFlow/EVA-Flow models through our evaluation code. The original and cleaned test sets share

Table 8: Median Recall and Precision COV under the 4-conformer protocol.

Environment	Setup	R-COV median $\uparrow$	P-COV median $\uparrow$
Vacuum	Pretrain+Unified	1.00	0.50
	Pretrain+Individual	1.00	0.50
	Unified	1.00	0.50
	Individual	1.00	0.50
	ETFlow (retrained)	1.00	0.50
Docking	Pretrain+Unified	1.00	0.50
	Pretrain+Individual	1.00	0.25
	Unified	1.00	0.50
	Individual	1.00	0.25
	ETFlow (retrained)	0.00	0.00
Solvation	Pretrain+Unified	1.00	1.00
	Pretrain+Individual	1.00	0.75
	Unified	1.00	1.00
	Individual	0.00	0.00
	ETFlow (retrained)	1.00	0.25
Packing	Pretrain+Unified	1.00	0.25
	Pretrain+Individual	1.00	0.25
	Unified	1.00	0.25
	Individual	1.00	0.25
	ETFlow (retrained)	0.00	0.00

Table 9: Pairwise-RMSD diversity ($\text{\AA}$) among the four generated conformers per molecule (Pre-train+Unified).

Environment	Mean $\uparrow$	Median	Std
Vacuum	0.915	0.903	0.620
Docking	2.227	1.828	1.508
Solvation	0.456	0.359	0.315
Packing	0.967	0.912	0.634

998/1,000 molecules, but differ in preprocessing and reference counts (up to 2,497 versus a cap of 30), which also changes the $2K$ sample budget. Comparisons within a matched block support model comparisons; values across datasets or implementation pipelines are diagnostic only and are not used for causal attribution.

F Physical Validity of Generated Conformers

Coverage and AMR measure geometric similarity to reference conformers but not energetic plausibility. To assess physical validity directly, we relax all 4,000 generated Vacuum conformers ($1,000 \text{ molecules} \times 4 \text{ conformers}$) with UMA (Wood et al., 2026), a large-scale pretrained molecular potential, and measure the

Table 10: Solvation evaluation with single- and multi-reference protocols.

Training	Protocol	Refs/mol	Samples/mol	R-COV $\uparrow$	R-AMR $\downarrow$	P-COV $\uparrow$	P-AMR $\downarrow$
Single conformer	4-conf.	1	4	0.936	0.157	0.693	0.431
Single conformer	2K	mean 37	2K	0.981	0.102	0.541	0.591
Multiple conformers	2K	mean 37	2K	0.962	0.111	0.518	0.612

Table 11: Non-learned reference methods under the 4-conformer protocol.

Environment	Method	R-COV $\uparrow$	R-AMR $\downarrow$
Vacuum	ETKDG+MMFF	0.409	1.18
	EVA-Flow (P+U)	0.747	0.468
Docking	ETKDG+MMFF	0.581	2.09
	EVA-Flow (P+U)	0.684	1.899
Solvation	ETKDG+MMFF	0.006	3.42
	GFN2-xTB+ALPB-water	0.006	3.38
	EVA-Flow (P+U)	0.949	0.153
Packing	ETKDG+MMFF	0.606	0.72
	EVA-Flow (P+U)	0.644	0.664

Table 12: Matched 2K evaluation (0.75 Å) on original and cleaned GEOM-DRUGS test sets. All rows within a block use identical evaluation code and test references.

Model	Training set	Test set	R-COV $\uparrow$	R-AMR $\downarrow$	P-COV $\uparrow$	P-AMR $\downarrow$
Published ETFlow checkpoint	Original	Original	0.549	0.766	0.339	1.057
ETFlow (retrained)	Original	Original	0.712	0.562	0.704	0.594
EVA-Flow	Original	Original	0.721	0.550	0.707	0.589
Published ETFlow checkpoint	Original	Cleaned	0.517	0.799	0.294	1.193
ETFlow (retrained)	Original	Cleaned	0.759	0.505	0.629	0.688
EVA-Flow	Original	Cleaned	0.763	0.503	0.632	0.686
ETFlow (retrained)	Cleaned	Cleaned	0.798	0.443	0.728	0.532
EVA-Flow	Cleaned	Cleaned	0.810	0.415	0.740	0.505

Table 13: UMA relaxation of EVA-Flow Vacuum conformers (4,000 conformers from 1,000 molecules).

Metric	Value
UMA relaxation converged	99.9% (3,995/4,000)
Mean ΔE after relaxation	0.302 kcal/mol
Median ΔE after relaxation	0.115 kcal/mol
Fraction with $\Delta E < 1$ kcal/mol	97.1%
Mean RMSD shift during relaxation	0.056 Å
Fraction with RMSD shift < 0.1 Å	84.2%

energy change and geometric shift upon relaxation (Table 13). The very small energy changes and atomic displacements indicate that EVA-Flow generates conformers that are already near local energy minima according to a state-of-the-art potential.

Table 14: Shared-molecule analysis after excluding all shared molecules from finetuning.

Pair (A/B)	n	own A	own B	cross A	cross B	generation shift
Solvation–Vacuum	142	0.29	0.90	3.35	3.25	3.32
Docking–Vacuum	558	0.95	0.81	1.17	1.19	1.16
Vacuum–Packing	487	0.38	0.43	0.64	0.59	0.62
Docking–Solvation	13	0.63	0.35	2.34	2.27	2.23
Packing–Solvation	23	0.14	0.14	1.48	1.47	1.50
Packing–Docking	64	0.36	0.46	0.51	0.37	0.40

Table 15: Environment-feature sensitivity (inference-time zeroing, no retraining). Thresholds: Solvation 0.5 Å, Packing 0.75 Å, Docking 2.0 Å.

Environment	Ablation	R-COV $\uparrow$	R-AMR $\downarrow$	P-COV $\uparrow$	P-AMR $\downarrow$
Solvation	Full	0.949	0.153	0.780	0.367
	NoSolvDesc	0.905	0.191	0.734	0.399
Packing	Full	0.644	0.664	0.410	1.073
	NoLattice	0.575	0.762	0.360	1.154
Docking	Full	0.684	1.899	0.512	2.471
	NoPocket	0.400	2.590	0.257	3.197

G Expanded Shared-Molecule Controls

Shared molecules are matched by canonical SMILES and aligned with consistent heavy-atom mappings before RMSD calculation. Table 14 reports the control in which all 1,201 molecules occurring in at least two finetuning environments are removed from every finetuning training set. For pair A/B , own A/B compares generations conditioned on each context with that context’s reference; cross A/B swaps the reference; and generation shift compares the two context-conditioned generations. All entries are mean heavy-atom RMSD in Å. Because the broad pretraining corpus includes Vacuum molecules, this control isolates finetuning exposure but does not guarantee pretraining decontamination.

The own-versus-cross pattern remains for pairs with well-separated references, but own-reference error increases and Packing–Docking remains weakly separated. These data reduce the concern that the original result is explained only by finetuning memorization; they do not eliminate possible pretraining exposure or establish causality from conditioning alone.

H Environment-Graph Sensitivity

To verify that the model actively uses each environment representation, we zero out domain-specific environment features at inference time and measure the performance change (Table 15). **NoSolvDesc** zeros the four solvation descriptors; **NoLattice** zeros the seven lattice/space-group parameters; **NoPocket** zeros the entire environment embedding for docking. All three ablations degrade performance, confirming that the model relies on each environment component, and the docking pocket is the most critical.

I Full Ablation Metrics

Figure 5 reports Recall COV for the original ablations. Table 16 provides their complete metrics across all four environments. NoBase is damaging within the original base–latent parameterization, but the matched-budget E4 control in Table 17 shows that coherently removing both the learned base and latent is competitive.

Table 16: Full ablation metrics (Unified setup, small model). Thresholds: Vacuum/Packing 0.75 Å, Solvation 0.5 Å, Docking 2.0 Å. Best per metric/environment in **bold**.

Metric	Setup	Vacuum	Docking	Solvation	Packing
R-COV $\uparrow$	EVA (Full)	0.740	0.674	0.930	0.640
	GCN	0.670	0.665	0.803	0.605
	Gaussian	0.405	0.237	0.822	0.312
	NoConsist	0.338	0.577	0.134	0.373
	NoBase	0.321	0.577	0.255	0.398
	NoFM	0.000	0.126	0.006	0.005
P-COV $\uparrow$	EVA (Full)	0.463	0.536	0.769	0.414
	GCN	0.396	0.490	0.497	0.384
	Gaussian	0.241	0.179	0.618	0.201
	NoConsist	0.184	0.379	0.102	0.206
	NoBase	0.177	0.376	0.140	0.220
	NoFM	0.000	0.057	0.003	0.002
R-AMR $\downarrow$	EVA (Full)	0.456	1.889	0.166	0.682
	GCN	0.623	2.153	0.285	0.714
	Gaussian	1.078	4.528	0.291	1.248
	NoConsist	1.108	2.221	0.896	0.990
	NoBase	1.113	2.160	0.790	0.963
	NoFM	2.136	3.536	1.735	2.214
P-AMR $\downarrow$	EVA (Full)	0.954	2.391	0.370	1.067
	GCN	1.117	2.937	0.593	1.117
	Gaussian	1.551	5.414	0.492	1.601
	NoConsist	1.569	2.941	0.921	1.400
	NoBase	1.560	2.711	0.902	1.363
	NoFM	2.502	4.020	2.145	2.581

Table 17: Matched-budget latent-free E4 control. E4 retains the environment embedding, fixed harmonic base, and FM decoder.

R-COV	Vacuum	Docking	Solvation	Packing
Matched full model	0.740	0.688	0.936	0.640
E4	0.751	0.730	0.986	0.633
E4 – full	+0.011	+0.042	+0.050	−0.007

J Data Exposure and Balanced Sampling

To test whether the Unified advantage stems from cross-environment transfer rather than data volume, we analyze total molecule exposure and run a balanced-sampling control. Table 18 (top) counts molecule samples over the stated 800 epochs. Unified starts from nominal 60/20/5/15% allocations of 100K samples, then caps each environment at its dataset size, yielding 91,216 effective samples per epoch; Individual models pass over their full dataset each epoch. Unified sees far fewer DRUGS samples than Individual (48.0M vs. 218.1M) yet outperforms it; for the small-data environments the Individual models already overfit (validation loss plateaus by epoch $\sim$ 30–50), so additional repetitions alone cannot explain the gains.

Table 18 (bottom) compares balanced (25/25/25/25%) against the default weighted sampling, both from the same pretrained checkpoint. Solvation and Docking see the *same* number of molecules per epoch under both schemes (capped at dataset size), yet Solvation drops 10.7% when the DRUGS allocation falls from 60%

Table 18: Top: total molecule exposure across training. Bottom: balanced vs. default sampling (Recall COV at the environment threshold).

Environment	Dataset	Unified/ep	Unified tot.	Indiv./ep	Indiv. tot.
Vacuum (DRUGS)	272,641	60,000	48.0M	272,641	218.1M
Packing (COD)	20,765	20,000	16.0M	20,765	16.6M
Solvation (AQM)	1,276	1,276	1.02M	1,276	1.02M
Docking (PDB)	9,940	9,940	7.95M	9,940	7.95M

Environment	Default (60/20/5/15)	Balanced (25/25/25/25)	Change
Vacuum (0.75 Å)	0.740	0.692	−6.5%
Docking (2.0 Å)	0.684	0.670	−2.0%
Solvation (0.5 Å)	0.949	0.847	−10.7%
Packing (0.75 Å)	0.644	0.643	−0.2%

Table 19: Inference noise-injection sweep on Vacuum (Pretrain+Unified checkpoint, SDE 50 steps): perturbing the harmonic prior by $\epsilon \cdot \mathcal{N}(\mathbf{0}, I)$.

ϵ	R-COV $\uparrow$	R-AMR $\downarrow$	P-COV $\uparrow$	P-AMR $\downarrow$
0.00 (default)	0.747	0.468	0.464	0.979
0.01	0.738	0.470	0.455	0.983
0.05	0.735	0.493	0.459	0.981
0.10	0.745	0.471	0.464	0.973
0.20	0.730	0.486	0.452	0.985
0.50	0.734	0.480	0.453	0.982
1.00	0.704	0.529	0.413	1.079

Table 20: Latent activation in the full and NoBase models.

Model	Active dims	KL/node (nats)	Pooled $\ \mu\ $	Environment-dependent
Full	0/128	$\approx 10^{-6}$	$\approx 10^{-3}$	No (cosine 1.000)
NoBase	3/128	up to ~ 3	≈ 0.21	Yes

to 25%, indicating that DRUGS data benefits Solvation through cross-environment transfer in the shared backbone, not through Solvation data volume.

K Train–Test Consistency of the Base Distribution

At inference the initial coordinates are drawn from the simple harmonic prior rather than the learned training-time base. The measured divergence at convergence is 0.036 nats (validation; 0.026 train). On 200 test molecules, the learned-to-prior eigenvalue ratio has median 1.0000 and all measured modes are within 1% of unity, with posterior mean approximately zero. A noise-injection sweep also degrades gradually (Table 19). These diagnostics indicate that the inference-time base mismatch is small. The large NoBase drop in the original parameterization is therefore consistent with an optimization interaction: its latent activates while the full-model latent collapses (Table 20), whereas the latent-free E4 model is competitive (Table 17). This evidence suggests, but does not causally prove, a training-scaffold mechanism.

Table 21: Latent-collapse diagnostics and intervention. “Random” samples $z \sim \mathcal{N}(\mathbf{0}, I)$; “zero” replaces the matched latent draw by zero. COV values use the environment-specific thresholds.

Environment	KL/node mean	KL/node median	Active dims	Posterior $ \mu $	R-COV random	R-COV zero	P-COV random	P-COV zero
Vacuum	1.5×10^{-6}	1.4×10^{-6}	0/128	8.2×10^{-5}	0.707	0.706	0.433	0.433
Docking	0.0476	1.4×10^{-6}	0/128	1.9×10^{-4}	0.661	0.661	0.497	0.497
Solvation	1.6×10^{-6}	1.5×10^{-6}	0/128	8.2×10^{-5}	0.803	0.809	0.580	0.591
Packing	1.6×10^{-6}	1.5×10^{-6}	0/128	8.2×10^{-5}	0.649	0.649	0.419	0.418

Table 22: One-factor-at-a-time loss sensitivity: change in Recall COV from unit weights.

Weight change	Vacuum	Docking	Solvation	Packing
latent-KL 0.1×	−0.018	−0.010	−0.020	−0.007
latent-KL 10×	−0.045	+0.018	−0.141	−0.030
base-KL 0.1×	−0.027	+0.004	+0.019	−0.008
base-KL 10×	−0.014	+0.032	+0.019	−0.015
consistency 0.1×	−0.018	+0.009	−0.064	−0.010
consistency 10×	−0.004	+0.028	+0.025	−0.001

L Latent Posterior-Collapse Diagnostics

We diagnose whether the variational latent z carries information in the Pretrain+Unified checkpoint. We compute the per-node latent KL, posterior statistics, and active dimensions (a dimension is active when the variance across data of its posterior mean exceeds 10^{-2}), then rerun generation after replacing sampled z by zero while preserving all subsequent random draws. Table 21 reports one matched diagnostic sampling seed rather than the five-seed means in Table 2; its purpose is the paired random-versus-zero intervention. The posterior is effectively collapsed: Vacuum, Packing, and Solvation have mean KL approximately 1.5×10^{-6} nats per node; Docking has a larger mean due to a small tail but the same near-zero median; no dimensions are active. For the same molecule across all six environment pairs, the pooled posterior mean has norm approximately 10^{-3} and is unchanged across contexts (cosine 1.000, relative $L_2 \leq 0.001$). Setting $z = 0$ leaves all metrics effectively unchanged. Thus the decoder primarily conditions on G and e , and we do not attribute its performance to information carried by z . The NoConsist degradation is instead interpreted as an optimization effect of that objective; E4 shows that z can be removed coherently.

M Loss-Weight Sensitivity

We vary one regularizer weight at a time by 0.1× and 10×, keeping the FM weight and training schedule fixed, and report the Recall-COV change from the unit-weight control (Table 22). In 21/24 cells, $|\Delta| < 0.045$. The larger changes are Vacuum with latent-KL 10× (−0.045), Solvation with consistency 0.1× (−0.064), and Solvation with latent-KL 10× (−0.141). Thus most cells are stable to large one-at-a-time changes, while Solvation is sensitive to particular latent-regularization settings. This experiment does not vary network capacity or interactions among weights.

N Environment-Encoder Equivariance

We replace the GCN environment encoder with a matched SE(3)-equivariant TorchMD encoder under the same controlled Unified training budget (Table 23). Accuracy improves for Docking, whose context carries genuine molecule-specific coordinates, while the encoder is neutral or worse for contexts dominated by topology, proxies, or scalar descriptors. This is an encoder comparison, not a claim that one architecture is uniformly superior.

For a separate joint rigid-rotation stress test over all 218 Docking test complexes, the GCN environment embedding changes by 1.33% on average (0.98% median relative error), with mean cosine similarity 0.99998

Table 23: Equivariant environment encoder minus matched GCN control. Positive Δ R-COV and negative Δ R-AMR favor the equivariant encoder.

Environment	Coordinate content of G_E	Δ R-COV	Δ R-AMR
Vacuum	zero-coordinate scaffold	-0.003	-0.011
Docking	physical pocket coordinates	+0.028	-0.082
Solvation	fixed proxy; scalar descriptors	-0.153	+0.139
Packing	RDKit proxy and lattice metadata	-0.006	+0.015

Table 24: Aquamarine test-reference sensitivity for a fixed model and sampling seed (0.5 Åthreshold).

Reference selection	R-COV $\uparrow$	R-AMR $\downarrow$	P-COV $\uparrow$	P-AMR $\downarrow$
c1 (default)	0.803	0.291	0.580	0.538
Minimum eSOLV	0.777	0.319	0.588	0.538
Deterministic random	0.803	0.292	0.584	0.538

Table 25: Aquamarine training-target sensitivity in matched 200-epoch runs.

Training-target selection	R-COV $\uparrow$
c1 (default rule)	0.580
Minimum eSOLV	0.561
Deterministic random	0.618

(minimum 0.99983). The equivariant encoder changes by approximately 10^{-5} and has cosine similarity 1.0000. This measures embedding stability, not end-to-end equivariance of generated coordinates.

O Aquamarine Conformer-Selection Sensitivity

The default protocol selects the stored c1 conformer (array index 0), not an energy-ranked ground state. To quantify reference sensitivity, we hold the trained model, molecule split, sampling seed, and four-conformer budget fixed while changing the test reference to either the minimum-eSOLV conformer or one deterministic random conformer per molecule (Table 24). We separately train matched 200-epoch models with each target rule (Table 25). These controlled runs use a smaller budget than the 800-epoch headline model and are compared only within their block.

P Molecule Visualization

The visualization of reference and generated conformers for molecules shared in the Packing-Docking environments.

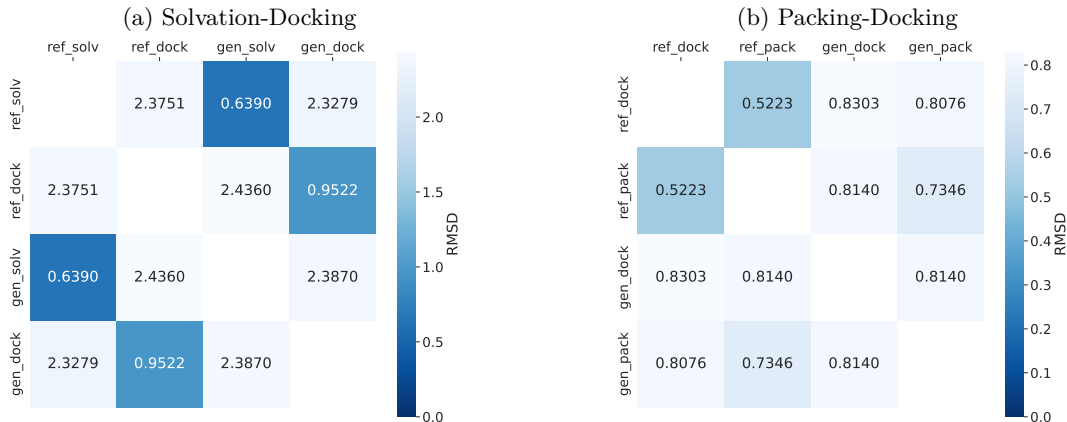


Figure 9: Pairwise heavy-atom RMSD (Å) between reference and generated conformers shared across: (a) Solvation-Docking, and (b) Packing-Docking. In each heatmap we report the mean RMSD after alignment for: $\text{ref_A} \leftrightarrow \text{ref_B}$, $\text{gen_A} \leftrightarrow \text{ref_A}$, $\text{gen_A} \leftrightarrow \text{ref_B}$, $\text{gen_B} \leftrightarrow \text{ref_A}$, $\text{gen_B} \leftrightarrow \text{ref_B}$, and $\text{gen_A} \leftrightarrow \text{gen_B}$ (A/B = the two tasks in the panel). Darker cells indicate lower RMSD (better agreement).

Table 26: Visualization of reference and generation conformations for the shared molecules in Packing-Docking. The molecules are aligned to the reference conformers of Docking environment.

